



< Previous





Next >

### 10.2.3 Deflating the problem

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Let's have a look at what happens to this matrix right here. Alright?

If everything is converging correctly, then what we expect to happen to this

matrix is that this last entry right here

ends up converging to the eigenvalue that is smallest in magnitude.

All of these values end up converging to

▶

2:53 / 2:53

▶ 2.0x

🔊

🔍

CC

“

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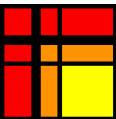
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Reading assignment

1/1 point (graded)



Read Unit 10.2.3 of the notes. [\[LINK\]](#)

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Homework 10.2.3.1

1/1 point (graded)  
Let  $A \in \mathbb{C}^{m \times m}$  be a Hermitian matrix and  $V \in \mathbb{C}^{m \times m}$  be a unitary matrix such that

$$V^H A V = \left( \begin{array}{c|c} A_{00} & 0 \\ \hline 0 & A_{11} \end{array} \right)$$

If  $V_{00}$  and  $V_{11}$  are unitary matrices such that  $V_{00}^H A_{00} V_{00} = \Lambda_0$  and  $V_{11}^H A_{11} V_{11} = \Lambda_1$ , are both diagonal, show that

$$\left( V \left( \begin{array}{c|c} V_{00} & 0 \\ \hline 0 & V_{11} \end{array} \right) \right)^H A \left( V \left( \begin{array}{c|c} V_{00} & 0 \\ \hline 0 & V_{11} \end{array} \right) \right) = \left( \begin{array}{c|c} \Lambda_0 & 0 \\ \hline 0 & \Lambda_1 \end{array} \right).$$

☒ done



Solution:

For a complete solution see the [notes](#).

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Answers are displayed within the problem

Homework 10.2.3.2

1/1 point (graded)  
Copy `SimpleShiftedQR.m` into `SimpleShiftedQRAlgWithDeflation` and modify it to add deflation:

```
function [ Ak, V ] = SimpleShiftedQRAlgWithDeflation( A, maxits,
illustrate, delay )
```

Modify the appropriate line in [Assignments/Week10/matlab/test\\_simple\\_QR\\_algorithms.m](#), changing `(0)` to `(1)`, and use it to examine the convergence of the method.

☒ done



Solution

For a complete solution, see the [notes](#).

< Previous

Next >

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